, at : all speeds again
pass . Those with speed satisfying

can also pass . Their speeds are

Times from to and to , respectively:

Time of arrival at :

At , shuts, hence

By and , being integral,

These particles enter the field with energies

Entry energies are, respectively,

At , shuts: no later entry to . By
, max entry energy is , giving greatest
speed . Using ,

Least energy gives least speed ; similarly,
we get

Determine the magnetic-field boundary
In the shown Cartesian field frame, entering at ,